

Event & Media Management Platform

Overview

Clubs and societies frequently organize events such as photoshoots, workshops, trips, competitions, cultural fests, and parties. During these events, a large number of photos and videos are generated. However, this media is usually scattered across multiple Google Drives, personal folders, and cloud storage links.

This creates major issues in organizing, searching, sharing, and managing event media efficiently.

The goal of this problem statement is to build a centralized Event & Media Management Platform where clubs, photographers, and members can upload, organize, access, and interact with media content seamlessly.

Problem Statement

Build a scalable web platform that allows photographers and organizers to upload event media and manage it efficiently. The platform should support:

- Event-wise media organization
 - Public and private access control
 - Social-media-like interactions
 - AI-powered image tagging and search
 - Cloud-based media storage
 - Personalized photo discovery using facial recognition
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Core Features

1) Event Management

Participants should implement:

- Create and manage events
- Event-wise albums
- Event descriptions and metadata
- Sorting by:
 - Event name

- Date
- Category

2) Media Upload System

The platform should support:

- Uploading photos and videos
- Bulk uploads
- Drag-and-drop upload support
- Media preview before upload
- Optimized storage and compression

3) Access Control & Authentication

The system should provide:

- Public Media: Accessible by everyone.
- Private Media: Accessible only to authorized club members.
- Suggested Roles:
 - Admin
 - Photographer
 - Club Member
 - Viewer

4) Social Features

The platform should include social interaction features such as:

- Like
- Comment
- Share
- Download
- Add to favourites
- Tag friends/users
- Real-time notifications

(Example notifications):

- Someone liked your photo
- Someone tagged you
- Someone commented on your upload

5) AI/ML Features

- Smart Image Tagging: Automatically generate tags such as:- mountains, beaches, sports, crowd etc.
- Advanced Search: Allow users to search by Event name, Tags, Upload date, and User name.

- Facial Recognition Feature

Users should be able to:

1. Upload a reference selfie
2. Detect photos containing that user
3. Display all matching photos in a separate personalized section

6) Cloud Integration

To increase scalability and industry relevance, participants should integrate cloud services.

- Suggested cloud services: AWS S3

7) Watermarking System The platform should support:

- Automatic watermarking during download
 - Dynamic watermark based on:
 - Club name
 - Event name
 - User role
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Bonus Features

Participants may implement additional innovative features such as:

- Infinite scrolling gallery
 - QR-based album sharing
 - Story/highlight feature
 - Collaborative albums
 - AI-generated captions
 - Analytics dashboard
 - Progressive Web App (PWA)
 - Offline caching
 - Image moderation system
 - Duplicate image detection
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Technical Expectations

Participants are expected to build a scalable and production-ready solution.

Expected Deliverables

Teams should submit:

Mandatory

- GitHub repository
- Working deployed project/demo
- Proper documentation/README
- Database schema
- Architecture diagram
- Presentation/PPT
- Demo Video

Optional

- API documentation
 - CI/CD pipeline
 - Docker setup
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Evaluation & Marking Criteria

Criteria	Weightage
UI/UX and Design	15%
Backend Architecture & APIs	15%
Authentication & Access Control	10%
Cloud Integration	15%

Criteria	Weightage
Media Management Features	15%
AI/ML Features	15%
Real-time Notifications	5%
Code Quality & Scalability	5%
Innovation & Bonus Features	5%

Judging Focus Areas

Judges will primarily evaluate:

- Problem understanding
- Scalability of architecture
- Real-world usability
- User experience
- AI/ML integration quality
- Cloud usage efficiency
- Feature completeness
- Creativity and innovation